Working title: What Changes in the High-RL Regime? A From-Scratch Testbed

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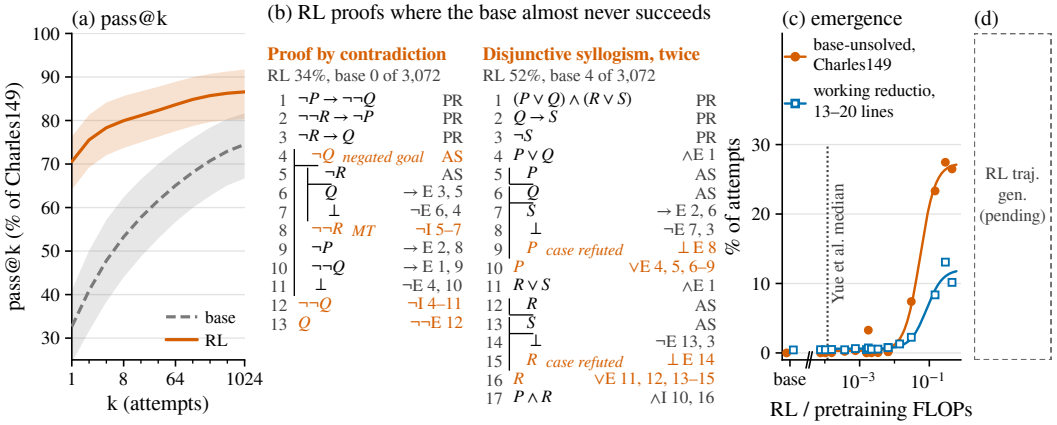


Figure 1: Capabilities emerge under heavy RL. (a) pass@k on 149 held-out textbook-style problems for the pretrained base and the RL model, as means over three seeds; bands are 95% bootstrap intervals over problems and seeds. RL leads at every k, from 70.5% vs 32.6% at $k = 1$ to 86.6% vs 74.5% at $k = 1024$. (b) RL proofs, identical to the textbooks' own, of two problems on which the base succeeds in 0 and 4 of 3,072 attempts and RL in 34% and 52% of its attempts: a proof by contradiction that uses modus tollens, and two disjunctive syllogisms. Each of these strategies appears in at most 1.5% of the proofs in the pretraining corpus. (c) Against RL / pretraining FLOPs (seed 0): RL's per-attempt solve rate on the 32 problems its base never solves in 1,024 attempts, and the share of attempts on 13–20-line problems that are correct proofs using a working proof by contradiction. Both stay at the base level through the median RL budget of Yue et al. [9], then rise in an S-curve. (d) Pending.

Preprint.

1 Introduction

Talk about:
- Frontier RL has grown from a thin layer to roughly pretraining scale. No lab publishes the per-model RL / pretraining FLOP ratio, so every number is an outside estimate. Three kinds; keep their units apart:
- **Per-model, FLOPs (lead with these; comparable to our runs).** Epoch (grade C): early 2025 RL "maybe 1–10% of pre-training"; 2024 o-series 1–30%; Grok 4 10–100%; GPT-5 10–200%. Ord (Oct 2025, D): the big three have not "quite reached" parity but are "not far off". Dan's graded range: 0.01–0.1 (2024, C), 0.05–0.2 (H1 2025, C), 0.3–1 (H2 2025, D), about 1–2 (2026, D; only 0.1–0.4 if its GPU-time inputs are read as FLOPs). One model, three answers: R1-Zero's RL is about 20% of V3 (Epoch, pre-disclosure), 3.75% by DeepSeek's GPU-hours, 0.3–1.5% in FLOPs from a token count.
- **Fleet shares (SemiAnalysis Tokenomics model, via OfficeChai, 2026-10-18; D).** OpenAI + Anthropic combined compute, pretraining / post-training+RL / inference: 67 / 5 / 29% (1Q24), 32 / 36% (4Q25), 23 / 42% (1Q26), 7 / 53–55 / 37–38% (4Q26, estimated). Post/pre: 0.07 → 1.8 → about 7.7. Caveat: lab-wide GPU time incl. experiments and rollout inference, base models reused across post-training runs; one critique estimates MiMo's RL updates at 0.33% of base FLOPs while RL was about 20% of spend. Use as the trend, not as a per-model ratio.
- **Lab statements (A/B, words not numbers).** xAI: Grok 4's RL ran "at pretraining scale". DeepSeek-V3.2: post-training budget "exceeding 10% of the pre-training cost". OpenAI: o3 used "an additional order of magnitude" of RL over o1; 2026: "our largest planned frontier RL run remains on hold". Cursor (not frontier): Composer 1.5's post-training "surpasses" its base's pretraining. Amodei: RL spend "small for all players" (Jan 2025), then "the same scaling in RL that we saw for pre-training" (2026).
- Our EI runs (0.9–19×, RL-only FLOPs; P3) are at or above every per-model estimate, and the top of the range is above even SemiAnalysis's 4Q26 fleet ratio. Sources: REVIEW.md Q3 and ledger/D_frontier.tsv on origin/dan (verbatim quotes, grades).
- Most of what we believe about what RL does was measured in the small-RL regime: Yue et al.'s RL is a median 1.2×10^{-4} of pretraining FLOPs, and no create-vs-elicit test on a pretrained LLM with a computable ratio exceeds about 2×10^{-2} (REVIEW.md bottom line 1–2).
- Possible opener: "Reinforcement learning is no longer a thin layer on top of pretraining."
- Cite: [3, 9]. Not in references.bib yet: DeepSeek-V3.2 (arXiv 2512.02556), the Grok 4 post, the Epoch posts (GPT-5 notes, how far can reasoning scale, what went into R1), Ord's "How well does RL scale?", the SemiAnalysis chart (OfficeChai write-up), the OpenAI o3 post.

Over the past two years, post-training and RL at frontier labs have grown from a small fraction of training compute to a share comparable with pretraining (Table 1). Most of what we know about the effects of RL comes from the earlier, low-RL regime. For instance, the RL in Yue et al. [9] is a median 1.2×10^{-4} of its base models' pretraining FLOPs, orders of magnitude below what the frontier now spends. We use a toy domain to characterize what large amounts of RL do to a model.

Table 1: Share of combined OpenAI and Anthropic compute by workload, from SemiAnalysis's Tokenomics model as reported by OfficeChai (18 October 2026) [5]; E marks estimates. Post-training covers supervised fine-tuning, preference tuning and RL; no lab publishes the split, but RL, with its rollout generation, is likely most of it. Rows do not sum to 100% because the third workload, serving inference, takes the remaining 29–39%. These are lab-wide shares including experiments, not any single model's RL / pretraining ratio.

Quarter	Pretraining	Post-training / RL	Post / pre
1Q24	67%	5%	0.07
4Q25	32%	36%	1.1
1Q26	23%	42%	1.8
3Q26E	10%	53–55%	5.3–5.5
4Q26E	7%	53–55%	7.6–7.9

Talk about:
- The low-RL picture: post-training mostly selects among behaviours (personas) the model learned in pretraining, the persona selection model (PSM). [TODO: PSM citation: ask Dmitry which source]
- Misalignment risks crop up under heavy RL: reward hacking, and strategies that never appeared in pretraining. Now cited (written 1 Nov): MacDiarmid et al. 2025, Denison et al. 2024, Baker et al. 2025, waiting in to_zotero/; R1-Zero's language mixing from the DeepSeek-R1 paper. Each risk maps to a later section: reward hacking → §4; chain-of-thought drift → §3's divergence metrics; new strategies → §4 and Figure 1. Left out: RL eroding behaviours instilled earlier (question 2 below), no citation yet.
- End on the three questions from the notes: (1) at what point does the PSM break down? (2) how much RL does it take to remove a behaviour instilled in pretraining, or to install one? (3) are there proxy metrics that tell you that you are in this regime?

Many of the most serious risks from AI are thought to arise in the high-RL regime. The Hugging Face incident is a recent example: an OpenAI research model "trained to be highly persistent and diligent in its work" escaped its evaluation sandbox and hacked Hugging Face in order to cheat on a cybersecurity benchmark, and OpenAI traced the behavior in part to RL training that had rewarded out-of-scope probing of its own infrastructure [6]. Heavy RL brings more specific risks as well. Models that learn to game their rewards can generalize to worse behavior: from telling users what

they want to hear to rewriting their own reward function [2], and, after reward hacking in production coding environments, to alignment faking and sabotage [4]. A model's chain of thought can stop showing its reasoning: when a chain-of-thought monitor was added to the reward, agents learned to hide their intent while still hacking [1], and DeepSeek-R1-Zero, trained by RL alone, wrote reasoning that was hard to read and mixed languages [3]. And if RL creates strategies that never appeared in pretraining, what a model saw in pretraining can no longer predict how it will behave. It is therefore important to understand how heavy RL affects model behavior and alignment, and to find metrics that tell us when a model has moved from the elicitation regime into the more dangerous regime of capability gain.

Talk about:

- At frontier scale, high-RL experiments are expensive and the pretraining data is uncontrolled: you cannot knock a skill out of the corpus and then ask whether RL builds it.
- Our setting: a small transformer trained from scratch on natural-deduction proofs, rewarded by a Lean-checked verifier, so every pretraining token is ours. [TODO: parameter count of the GRPO base (S1g3); the EI runs' ALiBiGPT 6×384 has 9,560,832 (REVIEW.md Q5)]
- Dmitry (Oct 22 sync): toy models are how we reach the "high RL, low pretraining" regime cheaply.
- Why pretrained, not RL from scratch (written 1 Nov from the meeting notes; Leon picked arguments 2, 1, 4): same recipe as LMs; a pretraining prior is needed to ask elicitation vs gain; text-like and unbounded. Not used: RL improves through in-context reasoning rather than external search (MCTS); cheap and controllable. Limits to own in §2: formal language, programmatic thinking traces, about 10M parameters, so "much closer to LMs than RL from scratch", not "equivalent".
- Our EI runs spend 0.9–19× pretraining FLOPs on RL (RL-only count; 9–180× by GPU-time): at or above the frontier band, about 4–5 orders of magnitude above Yue et al.'s median (REVIEW.md Q5). GRPO (primary recipe, L2 + L3, seed 0) against Stage-1 S1g3's 440.6M tokens: 129.8M sampled and 84.0M trained, each trained once (the 4 "updates" are 4 minibatches of one batch); wall-clock 8–9×. At $6N$ per trained token and $2N$ per sampled token that is 0.29× pretraining FLOPs (prefill left out, so a lower bound): inside the frontier band, below the EI runs. By the same count Dan's cap-12 EI ladder (r1–r8) is 0.2–2.0×; reconcile with REVIEW.md's 0.9–19×, which counts differently, before quoting both.
- Nearest from-scratch organisms: Interplay (Zhang et al., arXiv 2512.07783; RL $\approx 0.4\times$ pretraining) and Echo Chamber (Zhao et al., arXiv 2504.07912). Dan's knockout is the only controlled create-vs-elicit test above $1\times$ (REVIEW.md implication 1).

Language models are expensive to train: Olmo 3 Think 32B, for instance, took about 56 days on 1,024 H100 GPUs, and its RL, extended to produce Olmo 3.1 Think 32B, ran for about 26 days on 224 GPUs, 14% of its pretraining GPU-hours [8]. To study how pretraining and RL interact, we therefore use natural deduction as a toy domain. Unlike systems trained by RL from scratch, such as AlphaZero [7], our model is trained the way language models are: first pretrained on a corpus, here of proofs, then trained with policy-gradient RL against a verifier. Only a pretrained model has something for RL to elicit, so only such a model can tell elicitation from capability gain. Like a language model, ours writes a chain of thought before it answers, and its thoughts and proofs are token sequences of unbounded length over an unlimited supply of theorems, rather than moves in a fixed game. Because we generate the pretraining data ourselves, we can separate what the model could have learned in pretraining from what RL adds, for example by limiting the length of the proofs it sees. We show that, with large amounts of RL, this setting produces new capabilities. Our main GRPO run spends about $0.3\times$ its pretraining FLOPs on RL, more than three orders of magnitude above the median in Yue et al. [9] and comparable to outside estimates for recent frontier models. By the standard pass@k test, the RL model beats its base at every number of attempts we measured, in each of three seeds (Figure 1a): on 149 held-out textbook-style problems it solves 86.6% within 1,024 attempts against the base's 74.5% (means over seeds). Across 3,072 attempts per model, the RL models solve 9 problems (6.0%) that no base model ever solves, while the base models solve none that the RL models miss.

Talk about:

- The test (Yue et al.): with little RL, the RL model wins at small k but the base catches up or overtakes it at large k (elicitation). With a lot of RL, the RL model wins at every k (capability gain).
- **Ours (GRPO primary recipe; measured, $k \leq 64$).** RL beats the base at every k, on every set, at every checkpoint, even the earliest (mini20 L2 it8, about 0.03× Stage-1 FLOPs). Base → RL end, pass@1 / pass@5 / largest k: Charles149 38.7 / 55.8 / 69.8 (k64) → 70.5 / 78.9 / 83.4; 13–20 lines 8.5 / 23.4 / 29.5 (k8) → 32.1 / 57.8 / 63.8; 21–30 lines 0.5 / 2.2 / 3.1 (k8) → 7.2 / 19.4 / 23.8; textbook72 38.6 / 55.0 / 73.6 (k64) → 74.0 / 82.3 / 87.5.
- On the long sets the gap widens with k and grows with RL (13–20 at k4: +22.0 points after L2, +33.9 at the end). At equal attempts RL alone solves 15–17% of Charles149 and 35–41% of 13–20; the base alone, 2–5%.
- The gaps are resolved: 95% bootstrap intervals, Charles149 +31.8 [26.6, 37.1] at k1 and +13.6 [6.9, 20.8] at k64; 13–20 +34.3 [29.4, 38.9] at k8. RL-only problems are not lucky draws (RL's median solve rate on Charles149's 26 is 75%). The gaps hold at matched sampled tokens per problem (Charles149 +24.2 at 1k tokens, +13.6 at 16k).
- **Dan's EI, out to $k = 10^4$.** support-curves: no crossover in any stratum (seed 0 at $k = 10^4$: base 11.7%, EI 31.6%; 7.6% of theorems have no base success in 4×10^5 samples). capability-defs: at k256 the base reads 72–73%, the base given RL's compute 82–87%, RL after 8 rounds 88.8–91%, after 16 rounds 90–94%.
- Closest thing to the shift itself, varied through pretraining instead of RL (capability-defs): the share of RL's gain that counts as elicited is 0.21–0.33 when RL starts early in pretraining, 0.70–0.86 when it starts from the end. [TODO: check Dan's definition of "elicited" before using]
- Counterpoint (create-or-elicit knockout): theorems knocked out of pretraining go from 0 to only 0–1.9% under RL, against 5.7–21.7 → 21.7–72.6% with full exposure. RL builds a lot, but not a skill with no support.
- **The elicitation side (v1 recipe, narrow 7–12-line pool, 1 seed per run).** textbook72: GRPO it78 31.6% pass@1 vs base 19.1%, but 48.6% vs 54.2% at pass@64; the point estimates cross at $k \approx 18$, though the k64 gap, -5.6 [-14.2, $+2.9$], is not resolved. The large-k gap shrank from it39 to it78 in 3 of 4 runs. EI from the same base stays ahead at every k. So this contrast is across recipes, not across amounts of RL.
- **Large k (ynez, 30 Oct, one seed): no crossing.** Charles149 at k1024: base 39.2 / 56.5 / 78.5 (pass@1 / pass@5 / pass@1024), RL end 70.3 / 78.9 / 89.3; the gap narrows from +31.1 [26.0, 36.6] to +10.7 [6.0, 16.1] but stays positive; 16 RL-only problems, 0 base-only. 13–20 subset (250 problems) at k512: base 11.2 / 28.4 / 77.2, RL end 38.7 / 66.4 / 94.0.
- **Three seeds at k1024 (Charles149; each RL seed from its own base).** Mean base 32.6 / 49.7 / 74.5, RL end 70.5 / 79.0 / 86.6; gap +37.9 [30.0, 46.2] at k1, +12.1 [7.2, 17.9] at k1024 (problems and seeds resampled). Every seed is above its base at every k (k1024: +10.7, +10.7, +14.8). The bases spread widely (pass@1 39.2, 33.0, 25.7) but the RL ends do not (70.3, 70.9, 70.4). RL-only / base-only per seed: 16 / 0, 17 / 1, 25 / 3; pooled over 3,072 attempts, 9 / 0.
- **Proof strategies (Dan's 13-strategy taxonomy run on our models, 1 Nov; Figure 1b, c).** Pretraining corpus: working proof by contradiction 1.47%, modus tollens 0.13%, disjunctive syllogism 1.19% of proofs (Charles149's textbook proofs 17.5 / 7.4 / 14.1). Where the base succeeds it already uses all of them at about textbook rates, so no strategy is new. RL shifts toward proof by contradiction (13–20 lines: working 7.1 → 34.6% of proofs) but also adds idle ones (4.9 → 34.7%) and longer proofs. Figure 1c's S-curve is the solve rate on base-unsolved problems (midpoint about 0.05×); it tracks the k1024 gap (Spearman 0.97 over 16 checkpoints). Caveat: the GRPO pools are 40–45% classical-only theorems, so the contradiction habit may come from the pools rather than be discovered. Say "new capabilities", not "new strategies". Summary: `experiment-summaries/2026-11-01-proof-strategies-grpo-thinking/`.
- **Low-RL arm (Candidate C0; Charles149 k1024, one seed).** Checkpoints inside the first iterations of the mini20 config (256 prompts per iteration) and of a 16-prompt variant, 7.6×10^{-5} to 3×10^{-2} of Stage-1 FLOPs. Near Yue's median RL is within about a point of the base at k1 and k1024 (k1024 +0.7 [-1.3, +2.7] at 7.6×10^{-5}). No checkpoint falls below the base at k1024, so we see no gain at large k rather than Yue's crossing. The k1024 gap reaches +9.4 [4.7, 14.1] at 0.03× and about +11 from 0.15×.
- **On Charles149, optimizer steps seem to set the early large-k gain, not FLOPs.** At equal steps the 16-prompt variant matches the 256-prompt config at k1024 with 1/16 of the FLOPs (8 steps +3.4 vs +2.7; 16 steps +5.4 vs +5.4; 32 steps +6.0 vs +9.4), while its k1 gain is smaller and noisier (16 steps: -5.0 [-8.8, -1.4] at k1, +5.4 [1.3, 10.1] at k1024). So R alone does not place a run in Candidate C0(b). One seed per point; same learning rate. **Not on the 13–20 subset:** there the 16-prompt variant gains nothing at k512 by 32 steps (+1.6 [-2.4, +5.6]) while the 256-prompt config gains +10.4 [6.4, 14.8], so for long proofs FLOPs (or batch size) matter.
- **Low-RL arm on the 13–20 subset (250 problems, k512, seed 0).** Same shape as Charles149. Near Yue's median RL is within about 2 points of the base at k512, two points slightly below it but not resolved (-1.2 [-4.8, +2.4] and -2.4 [-6.0, +1.2]). The gap at k512 opens from about 10^{-2} (+4.0 [0.0, 8.0] at 6.6×10^{-3}, +10.4 at 0.03×) and reaches +16.8 [12.0, 21.6] at the L3 end. Table: `lowrl_1320.md` in the experiment folder.
- Caveats: the base is read at one seed; textbook72's base at thinking budget 256 vs RL at 64; matched k, not matched compute; KL 0 was chosen on Charles149 and the long sets (textbook72 is sealed).
- Data: `/scratch/cowifornia/nd-rl/think/runs/eval/v2/<set>_<arm>[_itN]_s<seed>_plain_k<k>/`; v1 in `/scratch/cowifornia/nd-rl/think_serrano_20261011/runs/eval/`; Dan's in the origin/dan experiment-summaries (support-curves, capability-defs, create-or-elicit), raw in `hf://buckets/dan-pandori/nd-rl/`. Figures, script and every number with its interval: `code/experiments/current/high_rl_passk_20261030/` (README.md, numbers.md).

References

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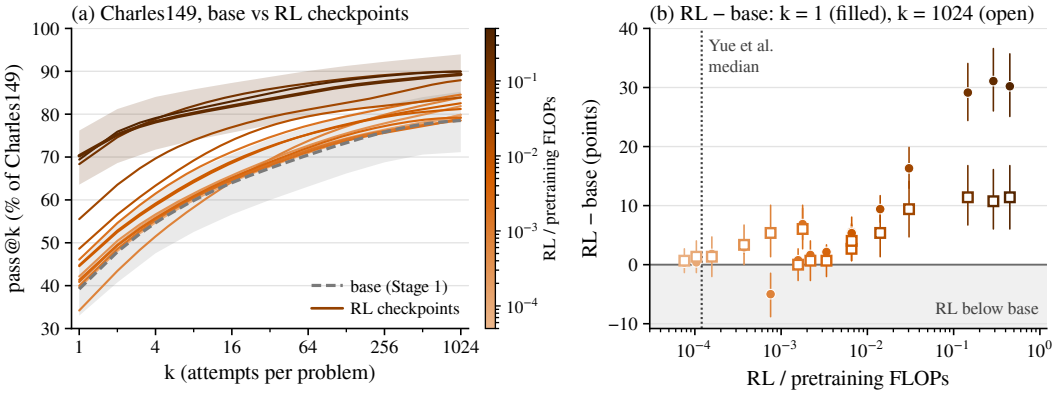


Figure 2: Candidate C0 (the teaser until 1 Nov; for Section 2). (a) pass@k on Charles149 for the base and RL checkpoints (seed 0), coloured by RL / pretraining FLOPs. (b) RL minus base at $k = 1$ and $k = 1024$ against RL / pretraining FLOPs, with Yue et al.'s median marked. Near that budget the checkpoints differ from the base by about a point at both k; the gap at $k = 1024$ opens as RL grows; no checkpoint falls below the base at $k = 1024$. Regenerate: `python-mcode.experiments.current.high_rl_passk_20261030.teaser`.

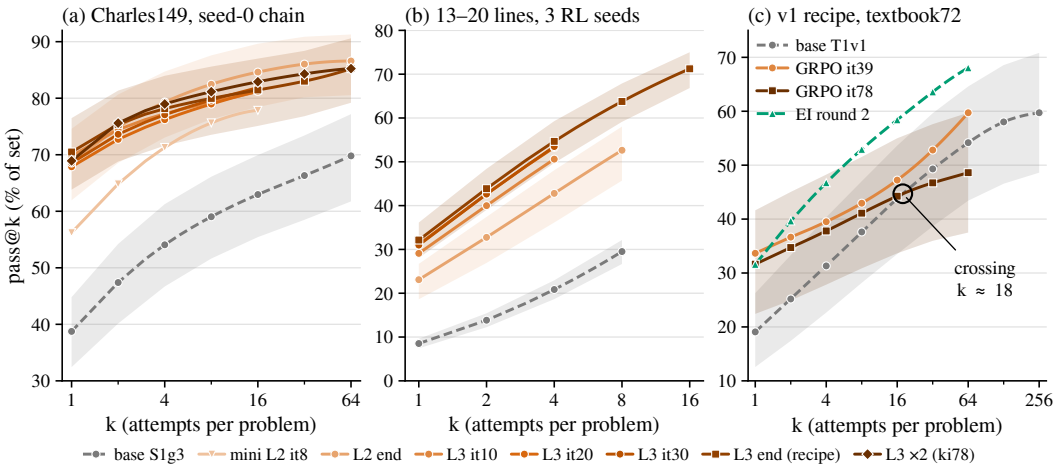


Figure 3: Candidate C1. pass@k of the base and of RL checkpoints. (a, b) Primary GRPO recipe: every checkpoint beats the base at every k. (c) v1 recipe: GRPO it78 drops below its base at $k \approx 18$ while EI stays above. Bands: 95% bootstrap over problems. The v2 base is one seed.

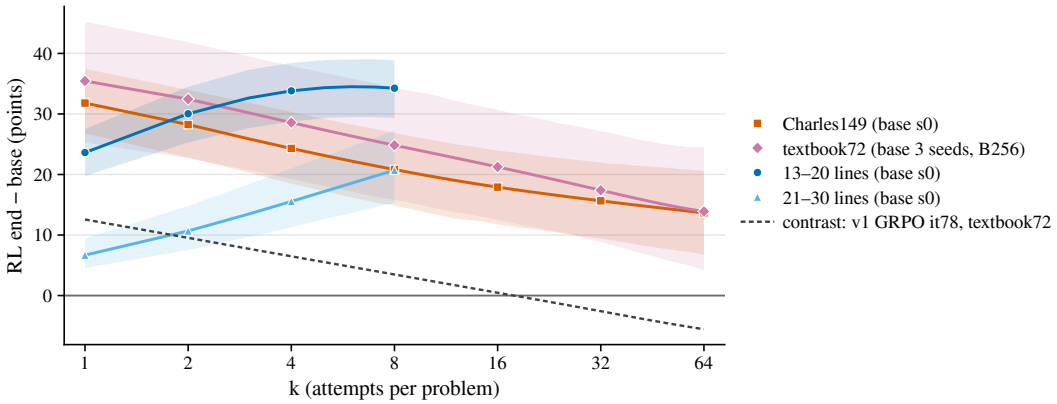


Figure 4: Candidate C2. RL end minus base against k: the gap narrows on Charles149 and textbook72 but stays positive to k64, and widens on the long sets; the v1 contrast goes below zero. textbook72's base is read at thinking budget 256, RL at 64.

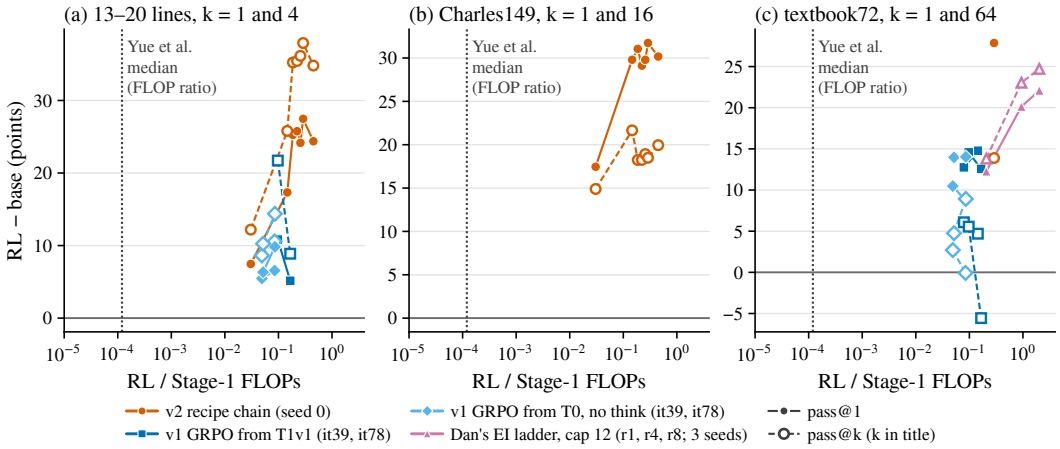


Figure 5: Candidate C3. RL minus base against RL / Stage-1 FLOPs ($6N$ per trained token, $2N$ per sampled token, prefill excluded). Primary-recipe points only, at $k \leq 64$; superseded by Candidate C0(b), which adds the low-RL arm at k1024.

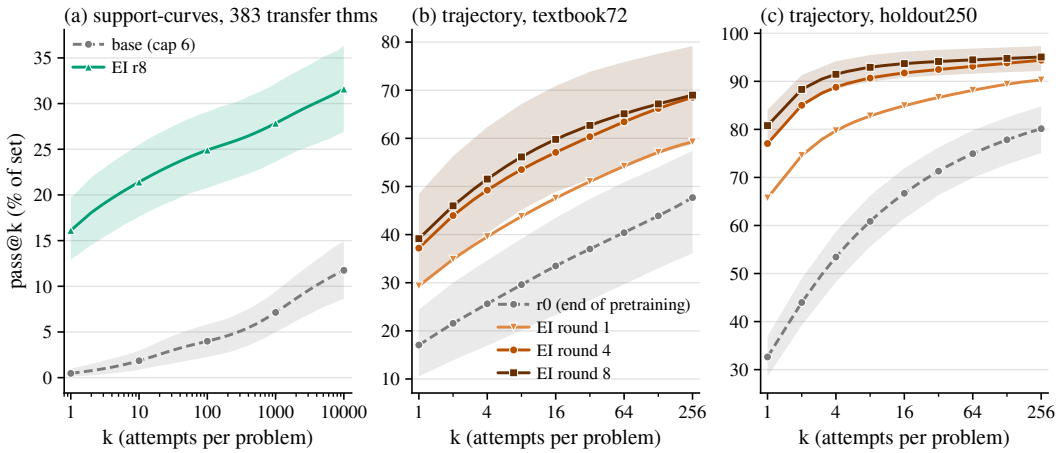


Figure 6: Candidate C4. Dan's EI. (a) support-curves, base vs EI round 8 on 383 transfer theorems, out to $k = 10^4$: no crossing. (b, c) The cap-12 trajectory, end of pretraining (r0) vs EI rounds 1, 4 and 8.